KAMAL NAYAN S

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PROFESSIONAL SUMMARY

Final-year Computer Science and Design student with strong foundations in Machine Learning, Data Structures, and Python. Experienced in building ML models including data preprocessing, training, and evaluation. Worked on real-world applications such as flight parameter prediction using neural networks and AI-based systems. Quick learner with adaptability to new technologies and a strong interest in building scalable, real-world solutions in dynamic environments.

TECHNICAL SKILLS

- **Programming Languages**: Java, Python, C, C++, R
- **Web Technologies**: HTML5, CSS3, JavaScript, React.js, Node.js, Vercel
- **Database Management**: MySQL, MongoDB
- **Tools/Platforms**: Git – version control, GitHub, VS Code

EDUCATION

Bachelor of Engineering – Computer Science and Design (On-going) 2022 – Present
MVJ College of Engineering (Autonomous), Bangalore, Karnataka

Pre-University (12th) – PCMC 2020-2022
East Point Group of Institutions, Bangalore, Karnataka

PROJECTS

Cognitive Companion Cross Platform

- Cross-platform application designed to support learning and cognitive assistance.
- Designed and supported backend workflows, ensuring smooth user interactions and resolving integration-related issues. Worked with structured user data for feature tracking and analysis to support adaptive system behavior.
- Analyzed and optimized user behavior data to improve system adaptability and feature performance.

Sportsight – Smart Equipment Identification System

- Developed a real-time object detection system to identify sports equipment (bats, balls, rackets, helmets) using computer vision techniques
- Implemented deep learning-based detection models (YOLO/OpenCV/CVlib) for accurate classification
- Performed image preprocessing and feature extraction using NumPy and OpenCV
- Integrated ESP32-CAM hardware with Python backend for real-time data capture and processing

Web Vulnerability Scanner

- Developed a full stack web application using Python Flask, HTML, CSS, and JavaScript.
- Designed a dashboard to visualize vulnerability reports and optimize scanning performance.
- Designed an analytics dashboard to display categorized vulnerability reports for improved decision-making.

CERTIFICATIONS AND ACHIEVEMENTS

- Smart IOT Innovation Challenges **[2nd Place]** – Sportsight-Smart Equipment Identification System [June 2024]
- AI Tools Workshop – be10x [May 2025]
- Figma Workshop – Google Developers Club
- Android Workshop – Google Developers Club

INTERNSHIP

HAL – Aircraft Research and Design Centre Division

- Developed a machine learning model using Multi-Layer Perceptron (MLP) to predict aircraft flight parameters
- Performed data preprocessing, model training, and error optimization using backpropagation
- Achieved improved prediction error of 0.006058250958877868
- Worked in a real-world R&D environment with system-level integration